Zepto Product Data Analysis Using SQL

1. Introduction

This project analyzes product data from Zepto, a quick-commerce grocery delivery platform.
The dataset contains information about product categories, pricing, discounts, stock availability, and product weights.

The goal of this analysis is to explore the dataset, clean inconsistencies, and generate insights such as:
• Best discount products
• Category revenue estimation
• Inventory analysis
• Price efficiency (price per gram)
• Stock availability patterns

The analysis was performed using SQL in PostgreSQL.

2. Database Setup

A database named ZEPTO was created to store and analyze the product dataset.

Table Name: zepto_analysis

Columns:
sku_id – Unique identifier for each product
category – Product category
name – Product name
mrp – Maximum Retail Price
discount_percent – Discount offered
available_quantity – Available stock quantity
discounted_selling_price – Selling price after discount
weight_in_gms – Product weight
out_of_stock – Indicates stock availability
quantity – Pack quantity

3. Data Exploration

Initial exploration was performed to understand the dataset.

• Count of rows to understand dataset size
• Viewing sample rows using LIMIT
• Checking for NULL values

• Identifying unique product categories
• Checking stock availability (in stock vs out of stock)
• Detecting products appearing multiple times (different SKUs or packaging sizes)

These steps help in understanding data quality and structure before analysis.

4. Data Cleaning

Data cleaning improves the accuracy of analysis.

Steps performed:
• Removed products with MRP = 0 since they represent invalid data.
• Converted prices from paise to rupees for easier interpretation.

Example transformation:
MRP = MRP / 100
Discounted Price = Discounted Price / 100

5. Data Analysis

Several analytical queries were executed to extract meaningful insights from the dataset.

1: Top 10 Best-Value Products Based on Discount
This query identifies products offering the highest discount percentage.

name	mrp	discount_percent
Dukes Wafl Dukes Waffy Chocolate Wafers	.00	51.00
Dukes Waffy Orange Wafers	45.00	51.00
Dukes Waffy Strawberry ...	45.00	51.00
Ceres Foods Fish Mustard ...	220.00	50.00
Ceres Foods Laal Maas Ins...	220.00	50.00
Ceres Foods Nalli Nihari In...	220.00	50.00
Chef's Basket Durum Whe...	160.00	50.00
Chef's Basket Durum Whe...	160.00	50.00
Chef's Basket Durum Whe...	160.00	50.00
Dukes Waffy Chocolate W...	150.00	50.00

2: High MRP Products That Are Out of Stock
Helps identify expensive products that are currently unavailable.

name	mrp
Patanjali Cow's Ghee	565.00
MamyPoko Pants Standar...	399.00
Aashirvaad Atta With Muti...	315.00
Everest Kashmiri Lal Chilli ...	310.00

3: Estimated Revenue per Category
Calculated using:
discounted selling price × available quantity

category	total_revenue
Fruits & Vegetables	10846.00
Meats, Fish & Eggs	20693.00
Biscuits	25007.60
Dairy, Bread & Bat...	55051.00
Beverages	55051.00
Health & Hygiene	64180.00
Home & Cleaning	122661.00
Packaged Food	224385.00
Ice Cream & Dess...	224385.00
Chocolates & Cand...	224385.00
Personal Care	270849.00
Paan Corner	270849.00
Cooking Essentials	337369.00
Munchies	337369.00

4: Premium Products with Low Discounts
Identifies products with MRP greater than ₹500 but discount less than 10%.

name	mrp	discount_percent
Dhara Kachi Ghani Mustar...	1250.00	8.00
Saffola Gold (Jar)	1240.00	0.00
Dhara Filtered Groundnut...	1050.00	1.00
Fortune Rice Bran Health ...	1050.00	1.00
Dhara Filtered Groundnut...	1050.00	0.00
Fortune Soyabean Oil	1005.00	0.00
Fortune Sunlite Refined S...	925.00	0.00
Surf Excel Matic Powder F...	810.00	7.00
Surf Excel Matic Top Load	720.00	9.00
Pedigree Puppy Dry Dog ...	690.00	6.00
Pedigree Dog Food Adult ...	660.00	7.00
Lizol Double Concentrate ...	650.00	8.00
Nestle Nestle Nan Pro Foll...	650.00	0.00
Dove Daily Shine Shampoo	645.00	5.00

5: Top Categories with Highest Average Discounts
Helps determine which categories provide the most promotional offers.

category	avg_discount
Fruits & Vegetables	15.46
Meats, Fish & Eggs	11.03
Chocolates & Cand...	8.32
Ice Cream & Dess...	8.32
Packaged Food	8.32

6: Price Per Gram Analysis
Calculated as:
discounted_selling_price / weight_in_gms
This helps determine best value products.

name	weight_in_gms	discounted_selling_price	price_per_gram
Aashirvaad Iodised Salt	1000	19.00	0.02
Onion	1000	21.00	0.02
Onion	3000	57.00	0.02
Shubh kart - Nirmal suga...	1160	28.00	0.02
Tata Salt	1000	24.00	0.02
Vicks Cough Drops Menthol	1160	20.00	0.02
Baby Potato	500	16.00	0.03
Beetroot	500	13.00	0.03
Carrot	500	15.00	0.03

7: Product Weight Classification
Products were grouped into:
Low (<1000g)
Medium (1000g–5000g)
Bulk (>5000g)

name	weight_in_gms	weight_category
Onion	1000	Medium
Tomato Hybrid	1000	Medium
Tender Coconut	58	Low
Coriander Leaves	100	Low
Ladies Finger	250	Low
Potato	1000	Medium
Lemon	200	Low
Watermelon	58	Low

8: Total Inventory Weight per Category
Calculated using:
weight_in_gms × available_quantity
This helps analyze warehouse inventory distribution.

category	total_weight
Meats, Fish & Eggs	48016
Biscuits	84431
Fruits & Vegetables	91794
Health & Hygiene	142904
Dairy, Bread & Bat...	143735
Beverages	143735
Personal Care	348187
Paan Corner	348187
Home & Cleaning	373161
Packaged Food	490797
Ice Cream & Dess...	490797
Chocolates & Cand...	490797
Cooking Essentials	1404654
Munchies	1404654

6. Key Insights

- Some products offer very high discounts making them attractive for customers.
- Several high-priced products are out of stock, suggesting strong demand.
- Revenue potential varies across product categories.
- Premium products often have lower discounts.
- Price-per-gram analysis helps customers identify better value products.
- Inventory weight analysis helps in logistics and warehouse management.

7. Conclusion

This project demonstrates how SQL can be used for real-world data analysis in the quick-commerce industry.

Through data exploration, data cleaning, and analytical queries, meaningful insights were generated about pricing strategies, discount patterns, and inventory distribution.

Such analysis helps companies optimize pricing, manage stock efficiently, and improve customer offerings.